



6428 - Towards self-consistent dust depletion, nucleosynthesis effects, and cosmic chemical evolution

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Atomic species in the interstellar medium (ISM) transition out of their gas phase mainly by depletion onto dust. Photoionization models are an essential investigation tool for studying cosmic plasmas. To study dusty regions of the ISM, it is necessary to have a self-consistent and complete dust depletion model. Abundance measurements of certain elements had previously found that in less dusty regions these values were greater than their reference abundances, indicating a positive depletion. Recent studies suggest that this results from supernovae enriching the ISM with alpha-elements. This phenomenon is called the nucleosynthesis effect. Here, I propose to improve the existing dust depletion model in the spectral simulation code Cloudy by incorporating the most recently published nucleosynthesis effects. The last update on the Cloudy dust depletions occurred before the recent findings on the effect of nucleosynthesis on dust depletions. To obtain state-of-the-art spectral simulations I will use the nucleosynthetic curves developed in the literature to improve the Cloudy dust depletion calculations so that the simulated spectra better reflect the chemical evolution of the cosmos. These improvements will be part of the 2025 release of Cloudy and will be openly available.

OBSERVING DESCRIPTION